JUMANJI XR – Survive the Adventure

An immersive VR board game for Meta Quest 2: roll the die with your own hands and cross a dangerous jungle board to reach the final temple.

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(The following describes the ideal version of the application we want to build. How much of it will be implemented by the end of the semester will be decided later. The minimum scope we commit to: the board, the die, token movement, one trap, one treasure and the victory system.)

User Story Map (2p)

Main components of the application and the cases they cover. Jumanji XR is a single-player game, so there is only one type of user.

1. **User 1 – The Player**

The player of Jumanji XR is a board-game fan or a casual VR user who owns a Meta Quest 2 and wants to play an adventure board game in a more immersive way than on a flat table or a screen. They may have little VR experience, so interactions must feel natural with bare hands (no controllers) and the game must be comfortable to play seated in a fixed position.

a. **Goal 1. The player wants to start the game quickly and understand how to play.**
   i. **Activity 1. To do this, in our app they will be able to open the game and move through the main menu using their hands.**
      1. Main menu with Play and Quit buttons above the table, pressed by poking them with a finger (hand tracking).
      2. A short “how to play” hint: throw the die, then move your token.
   ii. **Activity 2. Then they will be placed in a fixed position in front of the board.**
      1. The jungle board is fixed on a table in the middle of an old attic.
      2. The player has a fixed position (no movement) from which the whole board and the room are visible, with the board within arm's reach.

b. **Goal 2. The player wants to play a turn: roll the die and move their token forward.**
   i. **Activity 1. To do this, they will be able to roll the virtual die with their hands.**
      1. A physical die that can be grabbed and thrown (XR Interaction Toolkit + hand tracking).
      2. The result is detected automatically when the die stops moving.
      3. A die that falls off the board respawns.
   ii. **Activity 2. Then they will pick up their token and move it themselves to the corresponding tile.**
      1. The token can be grabbed with a hand gesture (pinch or grab) and carried over the board.

2. The player counts the tiles shown by the die and places the token on the target tile, which is highlighted as help.
3. The token snaps onto the tile when released nearby (tolerates imprecise tracking).
4. A token placed on the wrong tile returns to its previous tile.

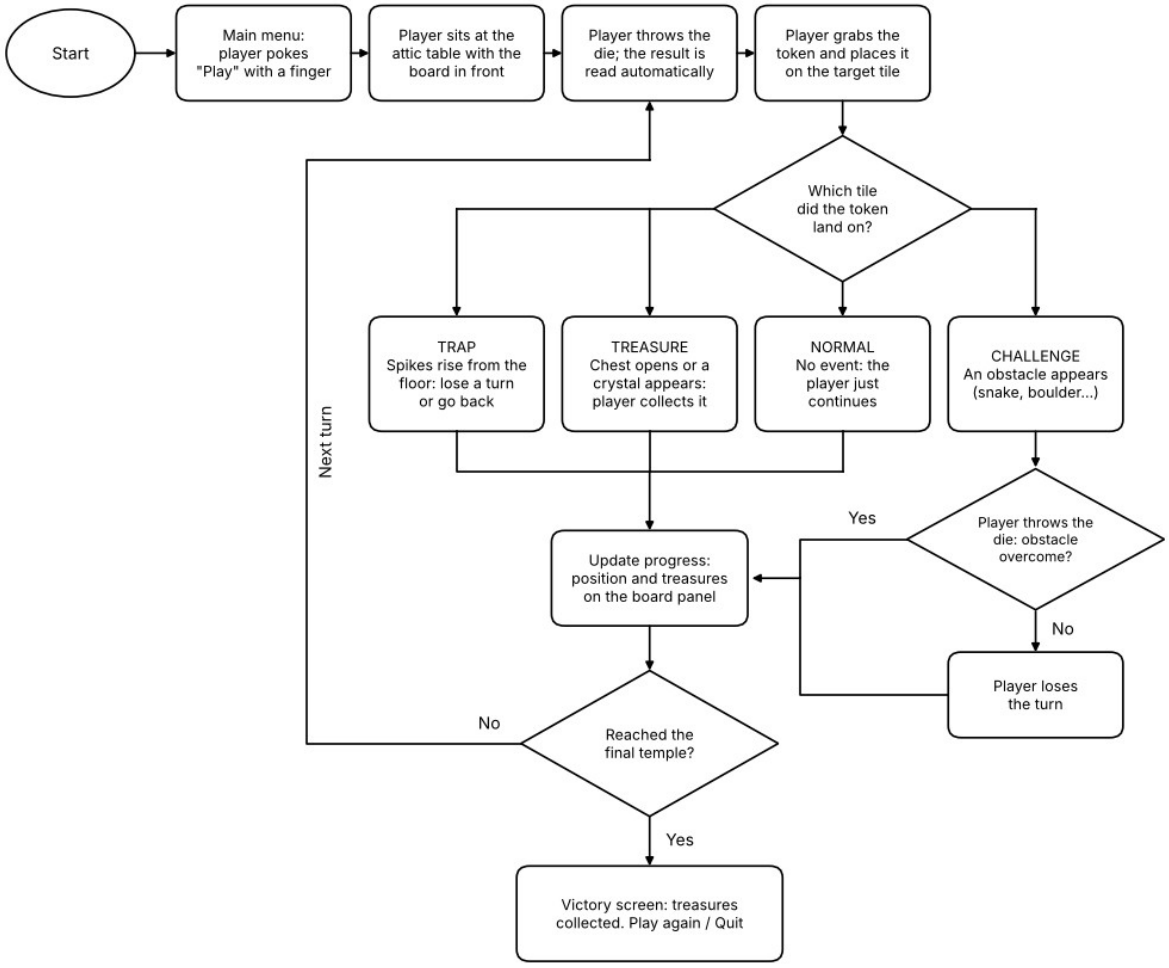
c. **Goal 3. The player wants to face surprises and rewards on the board.**
   i. **Activity 1. If the token lands on a trap tile, the tile opens and spikes come out.**
      1. Stone tile opens and spikes rise from the floor (animation and sound).
      2. Penalty (lose the next turn or go back a few tiles) with clear visual and audio feedback.
   ii. **Activity 2. If the token lands on a treasure tile, they will find a chest or a crystal.**
      1. A chest opens or a magic crystal appears on the tile.
      2. The player can pick up the treasure with their hand.
      3. The treasure counter increases.
   iii. **Activity 3. If the token lands on a challenge tile, they must overcome an obstacle (ideal version).**
      1. An obstacle appears: a snake, a rolling boulder or a fallen tree.
      2. The player throws the die: reaching a minimum number overcomes the obstacle; failing means losing the turn.

d. **Goal 4. The player wants to know how far they have got and to win the game.**
   i. **Activity 1. They will be able to check their progress at any time.**
      1. A panel on the board shows position, collected treasures and turn number.
   ii. **Activity 2. When the token reaches the final temple, they win.**
      1. The temple door opens with a light effect.
      2. Victory screen showing the treasures collected.
      3. Options to play again, go back to the main menu or quit.

e. **Goal 5. The player wants to feel inside an old attic where a dangerous jungle adventure comes to life, without discomfort.**
   i. **Activity 1. They will be able to enjoy an atmospheric attic environment.**
      1. Low-poly attic with wooden beams, a window, shelves, boxes, an old trunk and a rug.
      2. Warm window and lamp light, with dust floating in the air.
      3. Ambient attic sounds and adventurous music; the jungle board is the colorful focus of the room.
   ii. **Activity 2. They will be able to play comfortably for a whole session.**
      1. The player sits in a fixed position; there is no locomotion that could cause motion sickness.
      2. Stable 72 FPS on Meta Quest 2 (baked lighting, low-poly models).

Journey Map (2p)

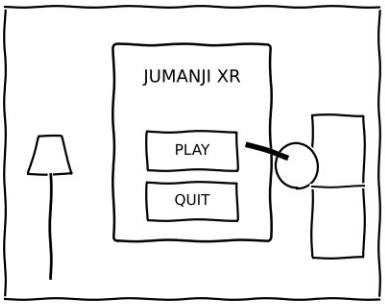
Journey map of the only way of using the application: playing a full game, from launching it to the victory screen. The flow is a single loop (roll the die, move the token, board event, update progress), so no additional journey maps are needed.



Wireframing/Storyboarding (3p)

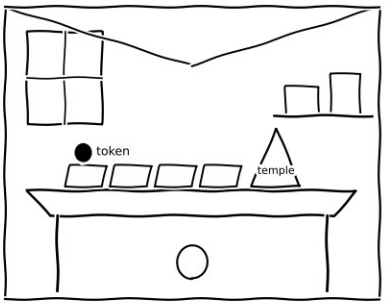
Key moments of the application, based on the main user flow. They are deliberately abstract sketches: visual design decisions (final models, colors and lighting) are not defined at this stage.

Title: JUMANJI XR Platform: Meta Quest 2 (hand tracking) Storyboard v1



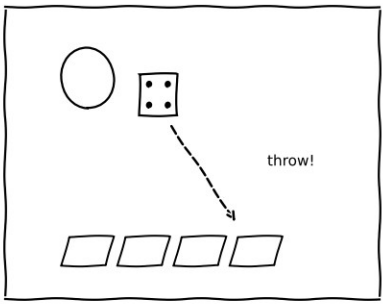
1. Main menu

A floating panel above the table. The player presses PLAY with a finger (hand tracking).



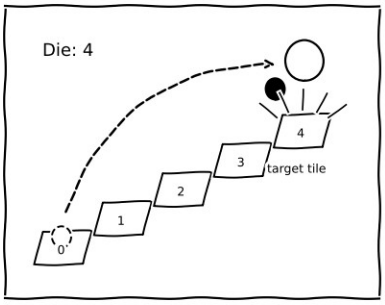
2. The player at the table

Seated in a fixed position in an attic, the player sees the jungle board on the table and the room around it.



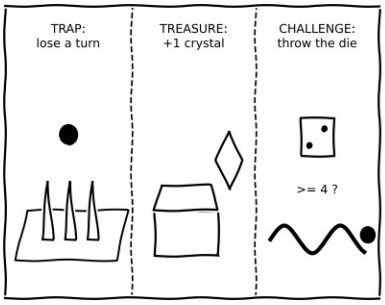
3. Roll the die

The player grabs the die with their hand and throws it onto the board.



4. Move the token

After the roll, the player picks up the token and places it on the tile that matches the die result.



5. Board events

Trap: spikes. Treasure: chest or crystal. Challenge: throw the die to beat the obstacle or lose a turn.



6. Victory

The token reaches the final temple. A victory screen shows the treasures collected.

Planning Asset Creation(3p)

Components we will need in the application. Convention used in the lists:

- Underlined: assets we will create from scratch (modeling, own design, own effects).
- Yellow highlight: assets we will generate (genAI tools, scripts).
- Plain text with a link: assets taken from other sources.

All 3D models follow the same low-poly style: flat shading, a limited color palette and no photographic textures.

3D

(Poly Pizza, Kenney, Unity Asset Store, Sketchfab, fab.com; Blender only for the token and the die)

- ➔ Player token: small adventurer figurine (Modeled in Blender)
- ➔ Die: one six-sided die with pips (Modeled in Blender)
- ➔ Board base and tiles: start, normal, trap, treasure and challenge (ProBuilder, in Unity)
- ➔ Attic room: walls, sloped roof and wooden beams (ProBuilder, in Unity)
- ➔ Final temple: low-poly stepped temple (Sketchfab https://sketchfab.com / Poly Pizza https://poly.pizza / Unity Asset Store https://assetstore.unity.com)
- ➔ Treasure chests, open and closed (Poly Pizza https://poly.pizza / Kenney https://kenney.nl/assets)
- ➔ Magic crystals (Poly Pizza https://poly.pizza / Kenney https://kenney.nl/assets); emissive material added in Unity
- ➔ Spike trap: spikes model (Poly Pizza https://poly.pizza / Unity Asset Store https://assetstore.unity.com); opening animation made in Unity
- ➔ Miniature palm trees, ferns and bushes for the board (Kenney Nature Kit https://kenney.nl/assets/nature-kit / Poly Pizza https://poly.pizza)
- ➔ Miniature rocks, stone pillars, ruins and stone idols for the board (Poly Pizza https://poly.pizza / Unity Asset Store https://assetstore.unity.com)
- ➔ Miniature hanging rope bridge for the board (Sketchfab https://sketchfab.com / Poly Pizza https://poly.pizza)
- ➔ Snake and other jungle animals for challenge tiles (Poly Pizza https://poly.pizza / Unity Asset Store https://assetstore.unity.com)
- ➔ Table where the board is placed and chair (Kenney Furniture Kit https://kenney.nl/assets/furniture-kit)
- ➔ Attic furniture and props: shelves, cardboard boxes, old trunk, rug, lamp, books (Kenney Furniture Kit https://kenney.nl/assets/furniture-kit / Poly Pizza https://poly.pizza)
- ➔ Window and decorations: picture frames, old toys, cobwebs (Sketchfab https://sketchfab.com / fab.com https://www.fab.com)

Graphics & 2D

- ➔ UI screens: main menu, in-game progress panel and victory screen (Own design)

- ➔ “Jumanji XR” logo / title (Own design)
- ➔ Icons: die, treasure counter, turn indicator (Own design)
- ➔ Menu and loading background art (AI generated)
- ➔ Sky seen through the attic window (Poly Haven HDRIs https://polyhaven.com/hdris / Unity Asset Store https://assetstore.unity.com)
- ➔ Wood and stone color references / tinted materials (Poly Haven https://polyhaven.com)

Sounds

(freesound.org, pixabay.com, elevenlabs, ...)

- ➔ Attic ambience loop: wind and creaking wood (Freesound https://freesound.org)
- ➔ Soft jungle ambience coming from the board (Freesound https://freesound.org)
- ➔ Background music: adventurous tribal drums (Pixabay Music https://pixabay.com/music / Freesound https://freesound.org)
- ➔ Die roll and die hitting stone (Freesound https://freesound.org)
- ➔ Token step on stone (Freesound https://freesound.org)
- ➔ Spike trap: stone sliding and spikes (Freesound https://freesound.org)
- ➔ Chest opening and crystal collect chime (Freesound https://freesound.org)
- ➔ Snake hiss for challenge tiles (Freesound https://freesound.org)
- ➔ Temple door opening and victory fanfare (Freesound https://freesound.org)
- ➔ UI button click (Freesound https://freesound.org)

VFX

- ➔ Crystal glow: emissive material and particles (Own design)
- ➔ Dust when the spikes come out of the board (Own design)
- ➔ Dust floating in the light of the window (Own design, Unity Particle System)
- ➔ Sparkle burst when a treasure is collected (Own design)
- ➔ Light beams and confetti when the temple opens (Own design)
- ➔ Dust when the die lands on the board (Own design)
- ➔ Soft glow of the lamp over the table (Own design, baked light)

Other assets

- ➔ Unity https://unity.com
- ➔ Meta XR SDK https://developers.meta.com/horizon/develop/unity
- ➔ XR Interaction Toolkit https://docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit@latest
- ➔ ProBuilder https://docs.unity3d.com/Packages/com.unity.probuilder@latest
- ➔ Blender https://www.blender.org
- ➔ Poly Haven https://polyhaven.com